B01 - ENGINEERING SQL TRANSITION

Course 2B DA-to-DE Bridge | Fast-track |

Why B01 exists

Course 1 already taught analyst SQL. B01 does not repeat that course. It converts those skills into engineering contracts: grain/cardinality, inspectable stages, deterministic windows, explicit time/KPI rules, repeatable quality diagnostics, independent reconciliation, plan awareness and maintainable handoff.

Exact study route

- 1. Run B01_POSTGRESQL_LAB_SETUP_RC1.sql in PostgreSQL.

- 2. Open B01_ENGINEERING_SQL_TRANSITION_LEARNER_BOOK_RC1.pdf and study B01-L01 only.

- 3. Complete P-B01-L01 in B01_ENGINEERING_SQL_TRANSITION_PRACTICE_RC1.sql.

- 4. Save SQL + validation + explain-back. Only then open B01-L01 in the protected practice review.

- 5. Close review and complete the changed retry. Repeat through B01-L08.

- 6. Complete Mini-Lab 1, then Mini-Lab 2.

- 7. Attempt Gate A independently. A clean pass ends B01.

- 8. Gate B/C stay sealed. A failed/contaminated Gate routes targeted repair; B/C require explicit fresh-full-retest assignment.

Video/source rule

Visual sources are support, not hidden prerequisites. The Alex SQL class is MySQL-based and is used only for visual concept refreshers. PostgreSQL 18 current documentation controls dialect-specific syntax and behavior. Every source has an internal course fallback.

B01 lesson sequence

ID	Lesson
B01-L01	Grain and cardinality contracts for engineering SQL
B01-L02	Staged transformations with CTEs and subqueries
B01-L03	Window patterns for ranking, latest-row and sequence logic
B01-L04	Conditional and date transformations with explicit business rules
B01-L05	Data-quality investigation queries
B01-L06	Reconciliation and validation SQL
B01-L07	EXPLAIN, scans and index awareness
B01-L08	Maintainable deterministic SQL handoff

Runtime note: live PostgreSQL execution remains a release QA check; static/fixture integrity is verified in this RC1.